COMPUTATIONAL SOCIOLOGY
Department of Sociology
University of North Carolina at Chapel Hill

Syllabus

Thomas Davidson

Fall 2026

CONTACT AND LOGISTICS

Course: SOCI 950-002

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Class meetings: Friday, 10:00 am–12:30 pm, 208 West Franklin Street, room WF-1020

Office hours: Wednesday, 10:00–11:00 am, 208 West Franklin Street, room 3111, or by appointment.

COURSE DESCRIPTION

This course introduces students to computational methods and their applications to sociological research. We will discuss the computational toolkit from the bottom up, beginning with the fundamentals of programming. Once these fundamentals are in place, we will explore different methods for collecting data from online sources including application programming interfaces, web-scraping, and online experiments. The remainder of the course will focus on computational methods for analyzing texts, images, and other types of data. The methods covered include word embeddings, topic modeling, and supervised machine learning. The final part of the class examines recent innovations in large language models and other forms of generative artificial intelligence, including how these systems work and how social scientists are using them to measure social phenomena. Throughout the course students will gain hands-on experience with these different techniques and an understanding of their applications in cutting-edge sociological research, including the strengths and weaknesses of these approaches.

COURSE GOALS AND LEARNING OBJECTIVES

By the end of this course, students will be able to:

- Understand and implement fundamental data structures and programming concepts.
- Collect observational data from online sources and build familiarity with online surveys and experiments.
- Apply a range of computational techniques to social science research questions, and understand how these techniques can complement conventional quantitative and qualitative approaches.
- Critically evaluate the use of new data sources and computational models, including methodological and ethical challenges.

PREPARATION

This course is designed for students without any experience using computational methods or advanced statistics. Students are expected to familiarize themselves with the R programming language, the RStudio computing environment, and RMarkdown. Information on learning resources is provided on the course website.

ASSESSMENT

Homework assignments and in-class exercises are designed to help students become familiar with the various methodological techniques covered in the course. Students will write an empirical paper over the course of the semester that will involve the collection of original data and/or implementation of analyses using one or more of the approaches covered in the course.

Grade breakdown

- Homework assignments: 30%
- Presentations: 20%
- Final project: 50%

Grading scale

The grading will follow the Graduate School's H/P/L/F scale.

The weighted components above are used to arrive at a final grade on this scale. Further information is available in the Graduate School Handbook.

Late work

Students should strive to complete work on time to avoid falling behind in the class. If you are unable to submit work by the due date, please be sure to contact me by email *in advance of the deadline* to discuss alternative arrangements.

ARTIFICIAL INTELLIGENCE (AI) USE

Generative AI tools are now a routine part of computational research, whether as coding assistants, as agents that carry out multi-step analytic tasks, or as measurement instruments, and I expect you to develop an

informed judgment about them. That judgment comes from using these tools enough to learn where they help and where they fail, and from studying them as objects of inquiry, which we will do throughout the semester.

Since the components of this course have different purposes, the level of AI use that is permitted varies across them. Coursework will involve a mix of the following, and I will state which applies to a given assignment or in-class exercise.

- **No AI:** Some exercises are designed to build the underlying skills and are to be completed without AI assistance. It is now trivial to use an AI model to generate code. It is critical to develop a solid grasp of the fundamentals so that you can evaluate and supervise any AI output. The same goes for your conceptual understanding of the methodological and theoretical content of the course. This applies to all course readings, as you are expected to read the texts and form your own judgment rather than relying on AI to summarize and synthesize.
- **AI-assisted work:** In some cases, you may use these tools as you would in your own research: to write and debug code, to work through unfamiliar packages and error messages, and to help revise your writing. You may not use them to generate text that you submit as your own prose.
- **AI as a method:** Some work will involve using generative AI as a measurement tool, for example to classify documents or annotate images. In these cases the question is not whether to use AI but whether the resulting measures are valid, and you will be expected to evaluate them appropriately.

In all cases, you are entirely responsible for what you submit: if a model produces an error and you do not catch it, it is your error. Verifying, testing, and being able to explain any code or claim you hand in is part of the work. *You will be asked to include a short statement in assignments describing how you used AI tools.* Please contact me if you are unsure whether a particular use is acceptable.

READINGS

There are weekly reading assignments for this course that you are expected to complete before each meeting. These readings include methodological texts, review articles, and examples of how sociologists and other social scientists apply computational techniques in their research. Further readings supplement the main texts and provide additional depth where needed.

Required texts

All required texts are available for free online and can be accessed via the links below.

- Salganik, Matthew J. 2017. *Bit by Bit*. Princeton University Press. ISBN: 0691158649
- Wickham, Hadley, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd edition. (*R4DS*). O'Reilly Media, Inc. ISBN: 1492097403
- Jurafsky, Daniel, and James H. Martin. 2026. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition.* 3rd edition draft.
- Silge, Julia, and David Robinson. 2017. *Text Mining with R: A Tidy Approach.* O'Reilly Media. ISBN: 1491981652

ADDITIONAL RESOURCES

The course website on GitHub contains all slides and resources used in class. We will cover how to use Git and GitHub in class.

POLICIES

Attendance

Regular attendance and active participation are expected. If you need to miss a class, please contact me in advance and we will discuss how to cover the material.

Correspondence

Please use your UNC email for any correspondence, and include the course code (950-002) in the subject line. I aim to reply to all emails sent during the work week within 24 hours, and may or may not reply on weekends and during holidays.

Diverse scholarly perspectives

Computational sociology is a rapidly changing field in which reasonable scholars disagree about methods, evidence, and interpretation. This course engages diverse scholarly perspectives to develop critical thinking, analysis, and debate, and the inclusion of a reading does not imply endorsement. Students are expected to read critically, to evaluate arguments on the basis of evidence and logic, and to engage with positions they may not share. Rigorous and respectful debate is a core part of the seminar.

Student conduct

All students are expected to adhere to University policy and follow the guidelines of the UNC Student Code of Conduct. In particular, students are expected to be aware of policies related to academic integrity. What constitutes cheating can vary from one activity to another: collaboration may be encouraged on some assignments and prohibited on others. Please see my guidelines for each activity, and if you are unsure, ask me to clarify. Additional information can be found at studentconduct.unc.edu.

Accommodations

Requests for accommodations are made through the University Compliance Office (see compliance.unc.edu). If you are eligible for accommodations, please let me know as early in the semester as possible so that I can support your work in this course.

Counseling and Psychological Services

Counseling and Psychological Services (CAPS) is the primary mental health provider for students, offering consultation and connection to clinically appropriate services. The Heels Care Network website provides access to the full range of mental health resources at Carolina. Students can call CAPS 24/7 at 919-966-3658 for immediate assistance.

Title IX and related resources

Any student impacted by discrimination, harassment, interpersonal (relationship) violence, sexual violence, sexual exploitation, or stalking is encouraged to seek support. Reports can be made to the University Compliance Office (UCO) or the University's Title IX Coordinator. Please note that I am designated as a Responsible Employee, which means I must report to the UCO any information I receive about these forms of misconduct. Confidential resources include CAPS, the University's Ombuds Office, and the Gender Violence Services Coordinators. Additional resources are available at safe.unc.edu.

Recording of class sessions

Students may not record or distribute recordings of class meetings without my prior written permission; those approved to record as an accommodation should contact me at the start of the semester.

COURSE OUTLINE

Week 1, 8/21

Introduction to Computational Sociology

Readings

- *R4DS*: Introduction, C1-4 (skim 1), skim 5, 6-8
- *Bit by Bit*, C1
- Lazer, David, et al. 2009. "Life in the Network: The Coming Age of Computational Social Science." *Science* 323(5915): 721–23.
- Golder, Scott A., and Michael W. Macy. 2014. "Digital Footprints: Opportunities and Challenges for Online Social Research." *Annual Review of Sociology* 40: 129–52.
- Edelmann, Achim, Tom Wolff, Danielle Montagne, and Christopher A. Bail. 2020. "Computational Social Science and Sociology." *Annual Review of Sociology* 46: 61–81.

Week 2, 8/28

Data Structures and Algorithms: From Networks to Agent-Based Models

Readings

- *R4DS*: C25-27, 16, 18-19.
- Bearman, Peter S., James Moody, and Katherine Stovel. 2004. "Chains of Affection: The Structure of Adolescent Romantic and Sexual Networks." *American Journal of Sociology* 110(1): 44–91.
- Backstrom, Lars, and Jon Kleinberg. 2014. "Romantic Partnerships and the Dispersion of Social Ties: A Network Analysis of Relationship Status on Facebook." In *Proceedings of the 17th ACM Conference on Computer Supported Cooperative Work & Social Computing*, 831–41.
- Bruch, Elizabeth, and Jon Atwell. 2015. "Agent-Based Models in Empirical Social Research." *Sociological Methods & Research* 44(2): 186–221.
- DellaPosta, Daniel, Yongren Shi, and Michael Macy. 2015. "Why Do Liberals Drink Lattes?" *American Journal of Sociology* 120(5): 1473–1511.

- Goldberg, Amir, and Sarah K. Stein. 2018. "Beyond Social Contagion: Associative Diffusion and the Emergence of Cultural Variation." *American Sociological Review* 83(5): 897–932.

Further reading

- Granovetter, Mark. 1973. "The Strength of Weak Ties." *American Journal of Sociology* 78(6): 1360–80.
- Breiger, Ronald L. 1974. "The Duality of Persons and Groups." *Social Forces* 53(2): 181–90.
- Burt, Ronald S. 2004. "Structural Holes and Good Ideas." *American Journal of Sociology* 110(2): 349–99.
- McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. "Birds of a Feather: Homophily in Social Networks." *Annual Review of Sociology* 27: 415–44.
- Block, Per, Christoph Stadtfeld, and Tom Snijders. 2019. "Forms of Dependence: Comparing SAOMs and ERGMs From Basic Principles." *Sociological Methods & Research* 48(1): 202–39.
- Macy, Michael, and Robert Willer. 2002. "From Factors to Actors: Computational Sociology and Agent-Based Modeling." *Annual Review of Sociology* 28: 143–66.
- Schelling, Thomas C. 1971. "Dynamic Models of Segregation." *Journal of Mathematical Sociology* 1: 143–86.
- Watts, Duncan. 1999. "Networks, Dynamics, and the Small-World Phenomenon." *American Journal of Sociology* 105(2): 493–527.
- Centola, Damon, and Michael Macy. 2007. "Complex Contagions and the Weakness of Long Ties." *American Journal of Sociology* 113(3): 702–34.
- Centola, Damon. 2015. "The Social Origins of Networks and Diffusion." *American Journal of Sociology* 120(5): 1295–1338.
- Park, Patrick S., Joshua E. Blumenstock, and Michael W. Macy. 2018. "The Strength of Long-Range Ties in Population-Scale Social Networks." *Science* 362(6421): 1410–13.

Week 3, 9/4

Collecting Digital Trace Data

Readings

- *R4DS*: C14, 17, 24
- *Bit by Bit*, C2, C6
- Askin, Noah, and Michael Mauskapf. 2017. "What Makes Popular Culture Popular? Product Features and Optimal Differentiation in Music." *American Sociological Review* 82(5): 910–44.
- Duxbury, Scott W., and Dana L. Haynie. 2021. "Shining a Light on the Shadows: Endogenous Trade Structure and the Growth of an Online Illegal Market." *American Journal of Sociology* 127(3): 787–827.
- Freelon, Deen. 2018. "Computational Research in the Post-API Age." *Political Communication* 35(4): 665–68.
- Li, Fumin, Yisu Zhou, and Tianji Cai. 2021. "Trails of Data: Three Cases for Collecting Web Information for Social Science Research." *Social Science Computer Review* 39(5): 922–42.

Further reading

- Fiesler, Casey, Nate Beard, and Brian C Keegan. 2020. "No Robots, Spiders, or Scrapers: Legal and Ethical Regulation of Data Collection Methods in Social Media Terms of Service." In *Proceedings of the Fourteenth International AAAI Conference on Web and Social Media*, 187–96.
- Haber, Jaren R. 2021. "Sorting Schools: A Computational Analysis of Charter School Identities and Stratification." *Sociology of Education* 94(1): 43–64.
- Tjaden, Jasper. 2025. "Web Scraping for Migration, Mobility, and Migrant Integration Studies: Introduction, Application, and Potential Use Cases." *International Migration Review* 59(3): 1367–84.
- Dahlke, Ross, Deepak Kumar, Zakir Durumeric, and Jeffrey T. Hancock. 2025. "Quantifying the Systematic Bias in the Accessibility and Inaccessibility of Web Scraping Content From URL-Logged Web-Browsing Digital Trace Data." *Social Science Computer Review* 43(5): 1071–86.

Week 4, 9/11

Online Experiments and Surveys

Readings

- *R4DS*: C9-11
- *Bit by Bit*, C3-5
- Salganik, Matthew J., Peter Sheridan Dodds, and Duncan J. Watts. 2006. "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market." *Science* 311(5762): 854–56.
- Kramer, Adam D. I., Jamie E. Guillory, and Jeffrey T. Hancock. 2014. "Experimental Evidence of Massive-Scale Emotional Contagion through Social Networks." *Proceedings of the National Academy of Sciences* 111(24): 8788–90.
- Munger, Kevin. 2017. "Tweetment Effects on the Tweeted: Experimentally Reducing Racist Harassment." *Political Behavior* 39(3): 629–49.
- Bail, Christopher A., Lisa P. Argyle, Taylor W. Brown, John P. Bumpus, Haohan Chen, M. B. Fallin Hunzaker, Jaemin Lee, Marcus Mann, Friedolin Merhout, and Alexander Volfovsky. 2018. "Exposure to Opposing Views on Social Media Can Increase Political Polarization." *Proceedings of the National Academy of Sciences* 115(37): 9216–21.

Further reading

- Wang, Wei, David Rothschild, Sharad Goel, and Andrew Gelman. 2015. "Forecasting Elections with Non-Representative Polls." *International Journal of Forecasting* 31(3): 980–91.
- Guess, Andrew M., Neil Malhotra, Jennifer Pan, Pablo Barberá, Hunt Allcott, Taylor Brown, Adriana Crespo-Tenorio, Drew Dimmery, Deen Freelon, Matthew Gentzkow, Sandra González-Bailón, Edward Kennedy, Young Mie Kim, David Lazer, Devra Moehler, Brendan Nyhan, Carlos Velasco Rivera, Jaime Settle, Daniel Robert Thomas, Emily Thorson, Rebekah Tromble, Arjun Wilkins, Magdalena Wojcieszak, Beixian Xiong, Chad Kiewiet de Jonge, Annie Franco, Winter Mason, Natalie Jomini Stroud, and Joshua A. Tucker. 2023. "How Do Social Media Feed Algorithms Affect Attitudes and Behavior in an Election Campaign?" *Science* 381(6656): 398–404.
- Rajkumar, Karthik, Guillaume Saint-Jacques, Iavor Bojinov, Erik Brynjolfsson, and Sinan Aral. 2022. "A Causal Test of the Strength of Weak Ties." *Science* 377(6612): 1304–10.

Week 5, 9/18

Fundamentals of Natural Language Processing

Readings

- *Text Mining with R*, C1, 3-5
- *Speech and Language Processing*, C5 (p. 131-136), C11 (p. 254-257).
- Evans, James, and Pedro Aceves. 2016. "Machine Translation: Mining Text for Social Theory." *Annual Review of Sociology* 42: 21–50.
- Nelson, Laura. 2020. "Computational Grounded Theory: A Methodological Framework." *Sociological Methods & Research* 49(1): 3–42.

Further reading

- Grimmer, Justin, and Brandon Stewart. 2013. "Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts." *Political Analysis* 21(3): 267–97.

- Bonikowski, Bart, and Laura K. Nelson. 2022. "From Ends to Means: The Promise of Computational Text Analysis for Theoretically Driven Sociological Research." *Sociological Methods & Research* 51(4): 1469–83.
- Denny, Matthew J., and Arthur Spirling. 2018. "Text Preprocessing for Unsupervised Learning: Why It Matters, When It Misleads, and What to Do about It." *Political Analysis* 26(2): 168–89.
- Stuhler, Oscar. 2022. "Who Does What to Whom? Making Text Parsers Work for Sociological Inquiry." *Sociological Methods & Research* 51(4): 1580–1633.

Week 6, 9/25

Word Embeddings and Computational Semantics

Readings

- *Text Mining with R*: C5.
- *Speech and Language Processing*, C6, pages 17-30.
- Hvitfeldt, Emil, and Julia Silge. 2020. *Supervised Machine Learning for Text Analysis in R*. "Chapter 5: Word Embeddings."
- Kozlowski, Austin, Matt Taddy, and James Evans. 2019. "The Geometry of Culture: Analyzing the Meanings of Class through Word Embeddings." *American Sociological Review* 84(5): 905–49.
- Rodriguez, Pedro L., and Arthur Spirling. 2022. "Word Embeddings: What Works, What Doesn't, and How to Tell the Difference for Applied Research." *The Journal of Politics* 84(1): 101–15.
- Boutyline, Andrei, and Alina Arseniev-Koehler. 2025. "Meaning in Hyperspace: Word Embeddings as Tools for Cultural Measurement." *Annual Review of Sociology* 51: 89–107.

Further reading

- Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. "Efficient Estimation of Word Representations in Vector Space." *arXiv*:1301.3781. Presented at the ICLR 2013 Workshop Track.
- Hamilton, William, Jure Leskovec, and Dan Jurafsky. 2016. "Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change." In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, 1489–1501.
- Rodman, Emma. 2020. "A Timely Intervention: Tracking the Changing Meanings of Political Concepts with Word Vectors." *Political Analysis* 28(1): 87–111.
- Gonen, Hila, and Yoav Goldberg. 2019. "Lipstick on a Pig: Debiasing Methods Cover up Systematic Gender Biases in Word Embeddings But Do Not Remove Them." In *Proceedings of NAACL-HLT*, 609–14.
- De-Arteaga, Maria, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. 2019. "Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting." In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 120–28.
- Stoltz, Dustin S., and Marshall A. Taylor. 2021. "Cultural Cartography with Word Embeddings." *Poetics* 88: 101567.
- Zhou, Di. 2022. "The Elements of Cultural Power: Novelty, Emotion, Status, and Cultural Capital." *American Sociological Review* 87(5): 750–81.
- Arseniev-Koehler, Alina, and Jacob G. Foster. 2022. "Machine Learning as a Model for Cultural Learning: Teaching an Algorithm What It Means to Be Fat." *Sociological Methods & Research* 51(4): 1484–1539.
- Best, Rachel Kahn, and Alina Arseniev-Koehler. 2023. "The Stigma of Diseases: Unequal Burden, Uneven Decline." *American Sociological Review* 88(5): 938–69.
- Rodriguez, Pedro L., Arthur Spirling, and Brandon M. Stewart. 2023. "Embedding Regression: Models for Context-Specific Description and Inference." *American Political Science Review* 117(4): 1255–74.

Week 7, 10/2

Topic Modeling

Readings

- *Text Mining with R*: C6.
- Blei, David. 2012. "Probabilistic Topic Models." *Communications of the ACM* 55(4): 77–84.
- Mohr, John, and Petko Bogdanov. 2013. "Introduction—Topic Models: What They Are and Why They Matter." *Poetics* 41(6): 545–69.
- Roberts, Margaret, Brandon M. Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David Rand. 2014. "Structural Topic Models for Open-Ended Survey Responses." *American Journal of Political Science* 58(4): 1064–82.
- Karell, Daniel, and Michael Freedman. 2019. "Rhetorics of Radicalism." *American Sociological Review* 84(4): 726–53.

Further reading

- DiMaggio, Paul, Manish Nag, and David Blei. 2013. "Exploiting Affinities between Topic Modeling and the Sociological Perspective on Culture: Application to Newspaper Coverage of U.S. Government Arts Funding." *Poetics* 41(6): 570–606.
- Eshima, Shusei, Kosuke Imai, and Tomoya Sasaki. 2024. "Keyword-Assisted Topic Models." *American Journal of Political Science* 68(2): 730–50.
- Hurtado Bodell, Miriam, Måns Magnusson, and Marc Keuschnigg. 2026. "Seeded Topic Models in Digital Archives: Analyzing Interpretations of Immigration in Swedish Newspapers, 1945–2019." *Sociological Methods & Research* 55(1): 120–56.
- Grootendorst, Maarten. 2022. "BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure." *arXiv*:2203.05794.
- Pham, Chau Minh, Alexander Hoyle, Simeng Sun, Philip Resnik, and Mohit Iyyer. 2024. "TopicGPT: A Prompt-Based Topic Modeling Framework." In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2956–84.

Week 8, 10/9

Supervised Machine Learning and Prediction

Readings

- Molina, Mario, and Filiz Garip. 2019. "Machine Learning for Sociology." *Annual Review of Sociology* 45: 27–45.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2021. "Machine Learning for Social Science: An Agnostic Approach." *Annual Review of Political Science* 24: 395–419.
- Hofman, Jake, Amit Sharma, and Duncan Watts. 2017. "Prediction and Explanation in Social Systems." *Science* 355(6324): 486–488.
- Salganik, Matthew, Ian Lundberg, Alexander Kindel, et al. 2020. "Measuring the Predictability of Life Outcomes with a Scientific Mass Collaboration." *Proceedings of the National Academy of Sciences* 117(15): 8398–8403.
- Brand, Jennie E., Jiahui Xu, Bernard Koch, and Pablo Geraldo. 2021. "Uncovering Sociological Effect Heterogeneity Using Tree-Based Machine Learning." *Sociological Methodology* 51(2): 189–223.

Further reading

- Mullainathan, Sendhil, and Jann Spiess. 2017. "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives* 31(2): 87–106.
- Watts, Duncan J. 2014. "Common Sense and Sociological Explanations." *American Journal of Sociology* 120(2): 313–51.
  - Turco, Catherine J., and Ezra W. Zuckerman. 2017. "Verstehen for Sociology: Comment on Watts." *American Journal of Sociology* 122(4): 1272–91.
  - Watts, Duncan. 2017. "Response to Turco and Zuckerman's 'Verstehen for Sociology.'" *American Journal of Sociology* 122(4): 1292–99.

No class 10/16, Fall Break

Week 9, 10/23

Supervised Text Classification

Readings

- Nelson, Laura K., Derek Burk, Marcel Knudsen, and Leslie McCall. 2021. "The Future of Coding: A Comparison of Hand-Coding and Three Types of Computer-Assisted Text Analysis Methods." *Sociological Methods & Research* 50(1): 202–37.
- Barberá, Pablo, Amber E. Boydstun, Suzanna Linn, Ryan McMahon, and Jonathan Nagler. 2021. "Automated Text Classification of News Articles: A Practical Guide." *Political Analysis* 29(1): 19–42.
- Miller, Blake, Fridolin Linder, and Walter R. Mebane, Jr. 2020. "Active Learning Approaches for Labeling Text: Review and Assessment of the Performance of Active Learning Approaches." *Political Analysis* 28(4): 532–551.
- Davidson, Thomas. 2025. "Hate Speech Detection and Bias in Supervised Text Classification." In *The Oxford Handbook of the Sociology of Machine Learning*, edited by Christian Borch and Juan Pablo Pardo-Guerra, 121–40. Oxford University Press.

Further reading

- Hanna, Alex. 2013. "Computer-Aided Content Analysis of Digitally Enabled Movements." *Mobilization: An International Quarterly* 18(4): 367–388.
- King, Gary, Patrick Lam, and Margaret E. Roberts. 2017. "Computer-Assisted Keyword and Document Set Discovery from Unstructured Text." *American Journal of Political Science* 61(4): 971–88.
- Duxbury, Scott W. 2023. "A Threatening Tone: Homicide, Racial Threat Narratives, and the Historical Growth of Incarceration in the United States, 1926–2016." *Social Forces* 102(2): 561–85.

Week 10, 10/30

Introduction to Large Language Models

Readings

- Manning, Christopher D. 2022. "Human Language Understanding & Reasoning." *Daedalus* 151(2): 127–38.
- Wankmüller, Sandra. 2024. "Introduction to Neural Transfer Learning With Transformers for Social Science Text Analysis." *Sociological Methods & Research* 53(4): 1676–1752.
- Brown, Tom B., et al. 2020. "Language Models Are Few-Shot Learners." In *Advances in Neural Information Processing Systems* 33, 1877–1901.

- Bender, Emily M, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–23.

Further reading

- Danescu-Niculescu-Mizil, Cristian, Robert West, Dan Jurafsky, Jure Leskovec, and Christopher Potts. 2013. "No Country for Old Members: User Lifecycle and Linguistic Change in Online Communities." In *Proceedings of the 22nd International Conference on World Wide Web*, 307–318.
- Jensen, Jeffrey L., Daniel Karell, Cole Tanigawa-Lau, Nizar Habash, Mai Oudah, and Dhia Fairus Shofia Fani. 2022. "Language Models in Sociological Research: An Application to Classifying Large Administrative Data and Measuring Religiosity." *Sociological Methodology* 52(1): 30–52.
- Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. "Attention Is All You Need." In *Advances in Neural Information Processing Systems* 30.
- Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding." In *Proceedings of NAACL-HLT*, 4171–86.
- Widmann, Tobias, and Maximilian Wich. 2023. "Creating and Comparing Dictionary, Word Embedding, and Transformer-Based Models to Measure Discrete Emotions in German Political Text." *Political Analysis* 31(4): 626–41.
- Bonikowski, Bart, Yuchen Luo, and Oscar Stuhler. 2022. "Politics as Usual? Measuring Populism, Nationalism, and Authoritarianism in U.S. Presidential Campaigns (1952–2020) with Neural Language Models." *Sociological Methods & Research* 51(4): 1721–87.

Week 11, 11/6

LLMs in Social Science Research

Readings

- Bail, Christopher A. 2024. "Can Generative AI Improve Social Science?" *Proceedings of the National Academy of Sciences* 121(21): e2314021121.
- Davidson, Thomas, and Daniel Karell. 2025. "Integrating Generative Artificial Intelligence into Social Science Research: Measurement, Prompting, and Simulation." *Sociological Methods & Research* 54(3): 775–93.
- Argyle, Lisa P, Ethan C Busby, Nancy Fulda, Christopher Rytting, and David Wingate. 2023. "Out of One, Many: Using Language Models to Simulate Human Samples." *Political Analysis* 31(3): 337–51.
- Park, Joon Sung, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. "Generative Agents: Interactive Simulacra of Human Behavior." In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- Argyle, Lisa P., Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. 2023. "Leveraging AI for Democratic Discourse: Chat Interventions Can Improve Online Political Conversations at Scale." *Proceedings of the National Academy of Sciences* 120(41): e2311627120.
- Ibrahim, Elida Izani, and Andrea Voyer. 2026. "Qualitative Research with LLM Chatbots: Technological Reflexivity for Interpretative Technology." *Qualitative Research* 26(1): 133–59.

Further reading

- Davidson, Thomas, and Daniel Karell, eds. 2025. *Special Issue: Integrating Generative AI into Social Science Research*. *Sociological Methods & Research* 54(3): 775–1242. Any article in this issue.
- Chae, Youngjin, and Thomas Davidson. 2026. "Large Language Models for Text Classification: From Zero-Shot Learning to Instruction-Tuning." *Sociological Methods & Research* 55(2): 501–67.

- Bisbee, James, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. "Synthetic Replacements for Human Survey Data? The Perils of Large Language Models." *Political Analysis* 32(4): 401–16.
- Jowsey, Tanisha, Virginia Braun, Victoria Clarke, Deborah Lupton, and Michelle Fine. 2025. "We Reject the Use of Generative Artificial Intelligence for Reflexive Qualitative Research." *Qualitative Inquiry*.
  - Ibrahim, Elida Izani, Laura K. Nelson, and Andrea Voyer. 2026. "Should Generative AI Be Used in Reflexive Qualitative Research?" *Qualitative Inquiry*.
  - Timmermans, Stefan, and Iddo Tavory. 2026. "AI and Qualitative Data Analysis: On the Vulnerability of Thematic Analysis and the Importance of Theorizing." *Qualitative Inquiry*.
- Hackenburg, Kobi, Ben M. Tappin, Luke Hewitt, Ed Saunders, Sid Black, Hause Lin, Catherine Fist, Helen Margetts, David G. Rand, and Christopher Summerfield. 2025. "The Levers of Political Persuasion with Conversational Artificial Intelligence." *Science* 390(6777): eaea3884.

Week 12, 11/13

Computer Vision and Multimodal Models

Readings

- Torres, Michelle, and Francisco Cantú. 2022. "Learning to See: Convolutional Neural Networks for the Analysis of Social Science Data." *Political Analysis* 30(1): 113–31.
- Gebru, Timnit, Jonathan Krause, Yilun Wang, Duyun Chen, Jia Deng, Erez Lieberman Aiden, and Li Fei-Fei. 2017. "Using Deep Learning and Google Street View to Estimate the Demographic Makeup of Neighborhoods across the United States." *Proceedings of the National Academy of Sciences* 114(50): 13108–13.
- Zhang, Han, and Jennifer Pan. 2019. "CASM: A Deep-Learning Approach for Identifying Collective Action Events with Text and Image Data from Social Media." *Sociological Methodology* 49(1): 1–57.
- Pessin, Léa, and Kevin Munger. 2026. "Beyond Text: Using AI-Generated Visual Conjoints to Study Gender and Housework Attribution." *Sociological Science* 13: 661–84.
- Davidson, Thomas. 2026. "Multimodal Large Language Models Can Make Context-Sensitive Hate Speech Evaluations Aligned with Human Judgement." *Nature Human Behaviour* 10(3): 514–30.
- Dietrich, Bryce J., and Melissa L. Sands. 2023. "Seeing Racial Avoidance on New York City Streets." *Nature Human Behaviour* 7(8): 1275–81.

Further reading

- Krizhevsky, Alex, Ilya Sutskever, and Geoffrey Hinton. 2012. "Imagenet Classification with Deep Convolutional Neural Networks." In *Advances in Neural Information Processing Systems*, 1097–1105.
- Jean, N., M. Burke, M. Xie, W. M. Davis, D. B. Lobell, and S. Ermon. 2016. "Combining Satellite Imagery and Machine Learning to Predict Poverty." *Science* 353(6301): 790–94.
- Buolamwini, Joy, and Timnit Gebru. 2018. "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification." In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, 77–91.
- Torres, Michelle. 2024. "A Framework for the Unsupervised and Semi-Supervised Analysis of Visual Frames." *Political Analysis* 32(2): 199–220.
- Hwang, Jackelyn, Nima Dahir, Mayuka Sarukkai, and Gabby Wright. 2023. "Curating Training Data for Reliable Large-Scale Visual Data Analysis: Lessons from Identifying Trash in Street View Imagery." *Sociological Methods & Research* 52(3): 1155–1200.
- Olah, Chris, Alexander Mordvintsev, and Ludwig Schubert. 2017. "Feature Visualization." *Distill* 2(11).
- Radford, Alec, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, et al. 2021. "Learning Transferable Visual Models From Natural Language Supervision." In *Proceedings of the 38th International Conference on Machine Learning*, 8748–63.
- Davidson, Thomas. 2024. "Start Generating: Harnessing Generative Artificial Intelligence for Sociological Research." *Socius: Sociological Research for a Dynamic World* 10.

- Zhang, Han, and Ryan Leung. 2026. "Joint Text-and-Image Clustering for Social Science Research." *Sociological Methodology* 56(1): 1–30.

Week 13, 11/20

Student presentations

- 12 minute presentation, plus Q&A.

No class 11/27, Thanksgiving Recess

Final paper

The final paper is the final assessment for this course.

Final paper due via email on 12/8 at 5pm